



Includes

Includes

Easy Web Exploitation picoCTF 2022 inspector

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Description

Can you get the flag?

Additional details will be available after launching your challenge instance.

This challenge launches an instance on demand.

Its current status is:

NOT_RUNNING

[Launch Instance](#)

Hints

1

Is there more code than what the inspector initially shows?

82,200 users solved



84% Liked



 picoCTF{FLAG}

[Submit Flag](#)

Author: The Analyst: Hyposelenia

Challenge: Includes

Category: Web Exploitation

Date: 02/21/26



I. Objective

The objective of this challenge is to **learn how websites split their code into multiple files (includes)** and how inspecting those included files can reveal hidden information such as flags.

II. Background

Modern websites are rarely written in a single HTML file. Instead, they use **includes**, meaning the website loads extra files such as:

- CSS files for design
- JavaScript files for behavior
- Sometimes additional HTML or server-side files

These files are **automatically loaded by the browser**, even if they are not visible on the screen.

In Capture The Flag (CTF) challenges, important information is often **hidden inside these included files**, assuming users will only look at the visible webpage.

III. Tool Used

- **Web Browser** - Used to access and interact with the challenge website
- **Browser Developer Tools (Inspect)** - Used to view page structure and cookies

IV. Methodology

1. Launched the challenge instance



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Its current status is:

NOT_RUNNING

Launch Instance

2. Accessed the provided web link after launching

On Includes

Many programming languages and other computer files have a directive, often called include (sometimes copy or import), that causes the contents of a second file to be inserted into the original file. These included files are called copybooks or header files. They are often used to define the physical layout of program data, pieces of procedural code and/or forward declarations while promoting encapsulation and the reuse of code.

Source: Wikipedia on Include directive

Say hello

3. Observed the landing page and clicked “Say Hello”, which displayed a message stating that “*This code is in a separate file*”

On Includes

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Source: Wikipedia on Include directive

Say hello

saturn.picocf.net:56189 says

This code is in a separate file!

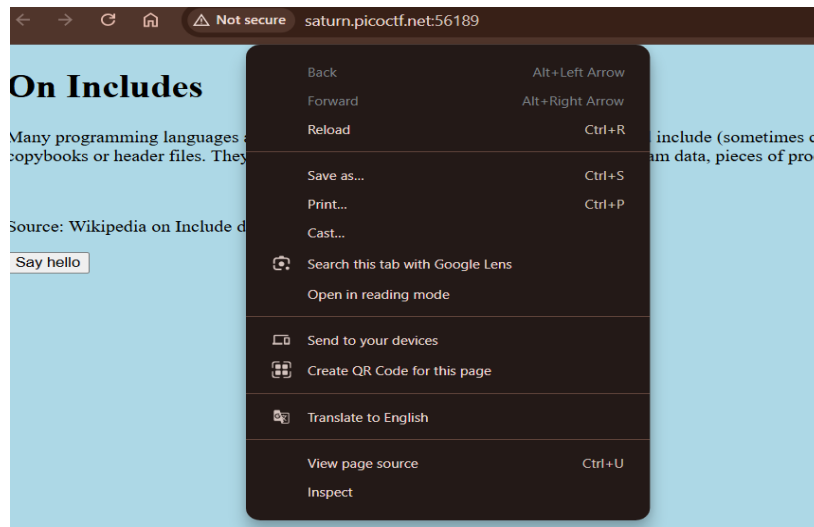
OK

second file to be inserted into the original file. These included files are called copybooks or header files. They are often used to define the physical layout of program data, pieces of procedural code and/or forward declarations while promoting encapsulation and the reuse of code.

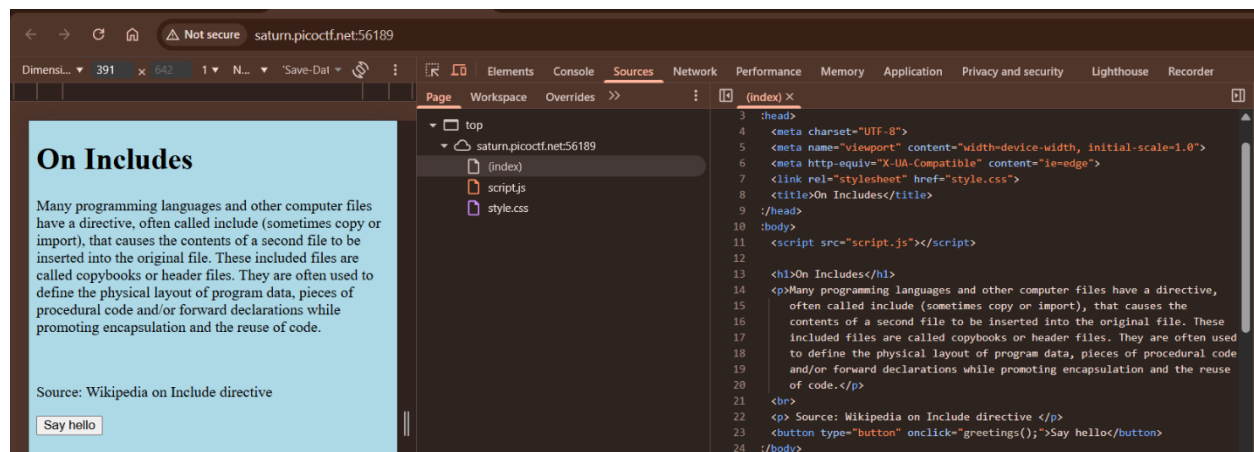
4. Since nothing suspicious appeared on the visible page, I opened **Inspect Developer Tools**



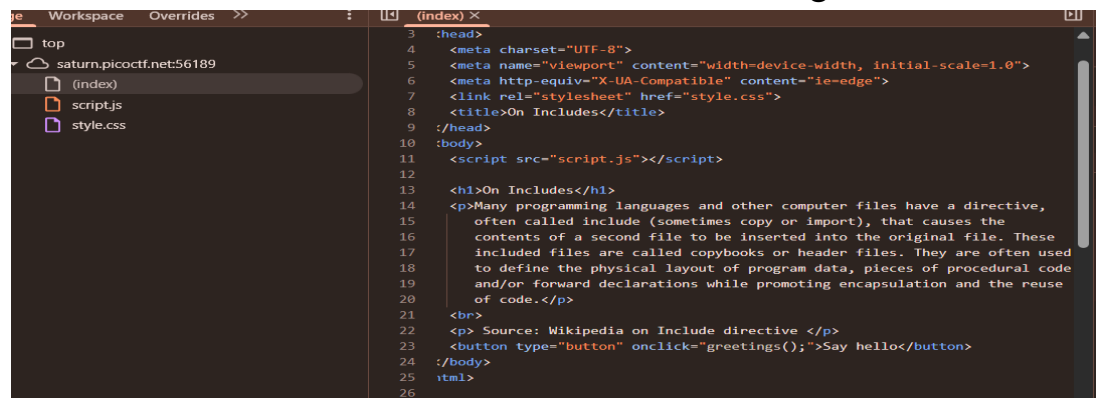
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5. Navigated to the **Sources** tab to view the structure of the website



6. Checked the main index.html file and found no flag





7. Navigated to the **CSS file** and discovered the **first fragment of the flag**, which I recorded

```
1 body {  
2   background-color: lightblue;  
3 }  
4  
5 /* picoCTF{1nclu51v17y_1of2_ */  
6 |
```

8. Navigated to the **JavaScript file** and discovered the **second fragment of the flag**, which I recorded

```
2  
3  
4 function greetings()  
5 {  
6   alert("This code is in a separate file!");  
7 }  
8  
9 // f7w_2of2_6edef411}  
10
```

9. Combined both parts to reconstruct the full flag

V. Result

picoCTF{1nclu51v17y_1of2_f7w_2of2_6edef411}

```
picoCTF{1nclu51v17y_1of2_f7w_2of2_6edef411}
```



VI. Explanation

What are “includes”?

Think of a website like a **coloring book**:

- The **HTML** is the drawing
- The **CSS** is the crayons
- The **JavaScript** is the instructions on how the drawing moves

Instead of putting everything on one page, websites **include** extra pages to keep things neat and organized.

What does “includes” mean in this challenge?

This website **included**:

- A CSS file
- A JavaScript file

Even though the flag was not visible on the webpage, the browser **still downloaded those files**, making them readable through Inspect.

Why did the flag appear in CSS and JavaScript?

Because:

- Included files are **public to the browser**
- Developers sometimes assume users won't inspect them
- CTF challenges intentionally hide clues there

This teaches that:

Anything sent to the browser is not secret.

Why is this important in cybersecurity?

In real websites:

- API keys
- Secrets
- Admin logic

are sometimes accidentally placed in included files.

Attackers:

- Inspect included files
- Extract sensitive data
- Abuse exposed logic

This challenge trains you to **never trust what you see on the screen alone**.



VII. Conclusion

The Includes challenge demonstrates that websites often rely on **external files** and that these files can contain sensitive or hidden information.

Through this challenge, I learned that:

- Inspecting included files is essential
- CSS and JavaScript can leak information
- Security should never rely on obscurity

This reinforces a core cybersecurity principle:

If the browser can load it, an attacker can read it.

— The Analyst: Hyposelenia